BRIMONIDINE

Not in children <2 yr

Allergic conjunctivitis common

CHEMICAL NAME 5-bromo-N-(4,5-dihydro-1H-imidazol-2-yl)quinoxalin-6-amine (tartrate)

CLASS Selective α2-adrenergic agonist (ophthalmic antiglaucoma)

TRADE NAMES Alphagan · Combigan (with timolol) · Mirvaso (rosacea gel) · Simbrinza (with brinzolamide)

THERAPEUTIC USES / INDICATIONS

- Open-angle glaucoma and ocular hypertension (alone or added).
- Facial erythema of rosacea (topical gel).

PHARMACODYNAMICS

- Stimulates **α2 receptors** → ↓ aqueous humour production and ↑ uveoscleral outflow → ↓ IOP; possible neuroprotective effect.

ADMINISTRATION

- **Eye drops:** 1 drop in affected eye(s); punctal occlusion; wait 5–10 min between drops; contact lenses out 15 min.
- **Gel (rosacea):** pea-sized amount once daily.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Allergic conjunctivitis/blepharitis (itchy red eyes) — common with long-term use.
- Dry mouth, fatigue, drowsiness.
- Headache, ocular stinging.

ADVERSE EFFECTS (SERIOUS)

- CNS depression, apnoea, bradycardia, hypotension in infants/young children.
- Hypotension, dizziness (adults).
- Follicular conjunctivitis.

DOSAGES

- 0.2%: 1 drop twice daily (three times daily in some regimens).
- 0.1–0.15% (Purite) formulations: 1 drop 2–3 times daily.

CONTRAINDICATIONS & CAUTIONS

- Children <2 yr (and caution <12 yr).
- Use with MAOIs, TCAs, mianserin.
- **Caution:** severe cardiovascular disease, depression, cerebral/coronary insufficiency.

KEY INTERACTIONS

- **MAOIs, TCAs:** contraindicated.
- **CNS depressants (alcohol, sedatives):** additive.
- **Antihypertensives:** additive hypotension.

PHARMACOKINETICS

- **Onset:** 1 h; peak 2 h. **Duration:** 12 h.
- Systemic absorption; crosses blood–brain barrier (CNS effects in infants).
- Metabolised in liver.

NURSING MANAGEMENT

- Check age before use.
- Assess for drowsiness, eye allergy.
- Monitor IOP and BP.

PATIENT EDUCATION

- May cause tiredness or drowsiness; take care driving.
- Report itchy, red, swollen eyelids (allergy).
- Never use in babies or young children.

KEY POINTS

- Ophthalmic α2 agonist.
- Allergic conjunctivitis; drowsiness.
- 1 drop BD–TDS.
- Contraindicated <2 yr (apnoea).